



Letter to the Editor

Water diffusion in silicate glasses under natural weathering conditions: evidence from buried medieval stained glasses

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Abstract

In this study, ion-microprobe analyses of four samples of buried medieval stained-glasses are used to demonstrate that water penetrates into the matrix of pristine glasses at low temperatures, thereby showing that glass alteration is not only a surface process. The diffusion coefficients of water determined from concentration profiles of hydrogen are found to be correlated with the bulk polymerization state of the glass. This observation is discussed with respect to glass structure and implies that ionic exchange between hydrogen and network modifying metal cations is the major process responsible for glass hydration.

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1. Introduction

Silicate glasses are extensively used as matrices for the confinement of toxic wastes coming from municipal, industrial and especially nuclear industry. To ensure the durability of such waste matrices, many studies were devoted to the understanding of mechanisms responsible for the deterioration of silicate glasses by water (e.g. [1–8]). Three main processes have to be considered during glass alteration: hydration, hydrolysis corresponding to the reaction of metal–oxygen bonds with water to form hydroxyl groups, and an ion exchange mechanism between network modifying cations and the protons or hydronium ions of the aqueous solution. For complex glasses, these three reactions can happen simultaneously [9]. When glass dissolution is not stoichiometric, a corrosion crust grows from the surface.

Three extreme hypotheses are usually assumed to explain the formation of the altered layer due to glass weathering: the first one considers incongruent glass dissolution leading to the precipitation of secondary products from a saturated solution [10,11]. The second assumption considers that the formation of an amorphous gel layer at the surface is due to a selective leaching of the glass [12,13], implying diffusion processes. The gel layer is in this case a heritage of the fresh glass and constitutes a residual layer. A third hypothesis considers that in situ silica recondensation plays a key role in the gel layer formation [14]. Glass hydration as well as gel layer formation are thus linked to diffusion processes in fresh or altered glass but mechanisms responsible for water penetration are not well known, especially if simple water permeation [15] or ionic exchange are involved. Based on a study of archaeological medieval stained-glasses, this paper shows that glass alteration is not only a surface reaction because water penetrates beyond the visible interface between fresh and altered glass and that, even at very low temperature, the apparently pristine glass is actually hydrated. Water profiles recorded in the pristine glass are interpreted in terms of diffusion

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profiles and the apparent diffusion coefficients are related to the bulk degree of polymerization of the glass.

2. Sample description

Four glasses unearthed in the course of archaeological excavations at two different sites within the Notre-Dame-du-Bourg church in Digne and at the bishop's palace in Rouen (France) were studied. They are dated from the 9th (Rouen) and 13th (Digne) centuries, respectively. They were altered during 800 (glasses from Digne) to 1100 years (glasses from Rouen) by natural humidity of the soil and humic acids. They belong to two different chemical families (Table 1): samples DV and DB from Digne and RVF from Rouen are rich in potassium (13–18 wt% K₂O), calcium (8–16 wt% CaO) and silica (50–59 wt% SiO₂), whereas RBa sample from Rouen is rich in sodium (18 wt% Na₂O), calcium (7 wt% CaO) and silica (68 wt% SiO₂). Each of them presents an adherent corrosion crust bordering the pristine glass (Fig. 1). This crust is highly hydrated (up to 20 wt% H₂O) and mainly consists of an alternation of hydrated silica gel and calcium phosphate phase layers [16]. Its thickness is a function of the bulk chemical composition of the glass (Table 2). It was demonstrated [17] that these archaeological samples were altered at constant volume, the free surface of the glass being conserved during weathering. The bulk degree of polymerization (NBO/T, [18]) is estimated for each glass by calculating the number of non-bridging oxygen per tetrahedron of network forming cation with the following equation:

$$\frac{\text{NBO}}{T} = \frac{1}{T} \sum_i n M_i^{n+},$$

where T is the number of network forming cations (Si⁴⁺, Ti⁴⁺, Al³⁺, Fe³⁺, P⁵⁺), M_i^{n+} is the number of network modifying cations (K⁺, Na⁺, Ca²⁺, Mg²⁺, Mn²⁺) after deduction of the number of Al³⁺ and Fe³⁺ charge

Table 1

Major elements in weight percent of oxide of the apparent pristine glass analysed by ICP-AES or EPMA for RBa sample

Wt%	DV	DB	RVF	RBa
SiO ₂	52.4 ± 1	50.0 ± 1	58.5 ± 1.2	70.0 ± 1.4
Al ₂ O ₃	1.7 ± 0.2	2.6 ± 0.2	2.2 ± 0.2	2.0 ± 0.2
Fe ₂ O ₃	0.7 ± 0.1	1.1 ± 0.1	0.5 ± 0.1	0.4 ± 0.1
MnO	0.7 ± 0.1	0.9 ± 0.1	0.8 ± 0.1	0.3 ± 0.05
MgO	5.1 ± 0.3	5.3 ± 0.3	5.0 ± 0.3	0.8 ± 0.05
CaO	15.5 ± 0.6	15.6 ± 0.6	12.9 ± 0.5	7.0 ± 0.3
Na ₂ O	1.1 ± 0.2	0.9 ± 0.1	1.4 ± 0.2	13.6 ± 2.2
K ₂ O	16.5 ± 0.7	17.8 ± 0.7	12.9 ± 0.5	1.0 ± 0.05
TiO ₂	0.2 ± 0.1	0.2 ± 0.1	0.4 ± 0.2	0.1 ± 0.05
P ₂ O ₅	4.1 ± 0.4	4.2 ± 0.4	2.9 ± 0.3	0.1 ± 0.05
Total	98.0 ± 3.7	98.6 ± 3.6	97.5 ± 3.6	95.3 ± 4.45
NBO/T	1.26	1.24	1.01	0.59

Total is not equal to 100% because of the presence of non-measured (coloring elements) or volatile elements (e.g. H₂O, Cl). Values of bulk degree of polymerization NBO/T is also reported for each sample.

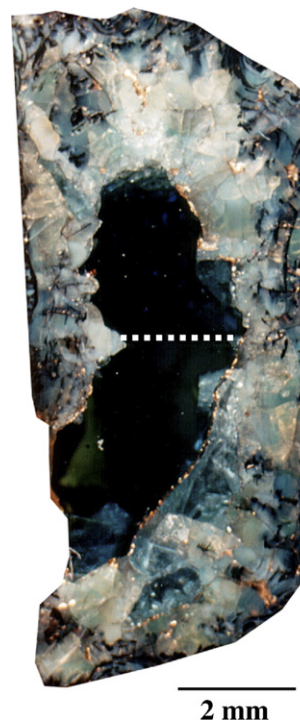


Fig. 1. Cross-section of DB sample (blue potassic stained-glass) buried in the soil during 800 years. The white dashed line shows the ion microprobe profile presented in Fig. 2. The 'fresh' glass (dark) is bordered by a thick altered layer (white).

compensators, and n is the electric charge of the i th cation. Because medieval glasses were made under atmospheric conditions, thus in contact with atmospheric oxygen, iron is assumed to be under 3+ oxidation state. Phosphorus is considered to be a network forming cation in agreement with [19]. NBO/T values calculated from our samples are reported in Table 2 and show that the sodic glass (RBa), richer in silica, is more polymerized than the potassic glasses which have a lower silica content. Moreover, comparison of altered layer thickness with NBO/T shows that the more the glass is polymerized, the less it is leached. These samples and the features of alteration are extensively described in [16,20].

3. Analytical techniques

Analyses of hydrogen were performed with ion microprobe CAMECA Ims-3f (modified CRPG). The samples were polished to 1 μm, ultrasonically cleaned in pure alcohol and baked at 40 °C during one night before coating with gold. The intensities on peaks ¹H⁺ and ³⁰Si⁺ were recorded with the following analytical conditions: O[−] primary beam without primary filter with an intensity of 2–5 nA and a size of 5–10 μm at a mass resolution of 1000 and with an energy filtering of −80 ± 10 V. A liquid nitrogen cold trap in the source was used to limit H₂O background. These experimental conditions allow a statistical error for H/Si ratio better than 5% [21]. The profiles are carried out on cross-sections, perpendicular to the fresh/

Table 2

Parameters used for the calculation of apparent hydrogen diffusion coefficients

Sample	DV	DB	RVF	RBa
Thickness of altered layer (μm)	1000 ± 200	1100 ± 200	450 ± 100	30 ± 6
Approximate duration of alteration (year)	760 ± 40	760 ± 40	$1,145 \pm 5$	1145 ± 5
Distance of water diffusion (μm)	800	800	600	50
a (m/s)	$4.2 \pm 1 \times 10^{-14}$	$4.4 \pm 1 \times 10^{-14}$	$1.3 \pm 0.3 \times 10^{-14}$	$8.6 \pm 1.8 \times 10^{-16}$
C_1 (mol/cm ³)	$5.41 \cdot 10^{-4}$	$4.40 \cdot 10^{-4}$	$7.58 \cdot 10^{-4}$	$1.28 \cdot 10^{-4}$
ppm H ₂ O	1987	1615	2786	480
C_2 (mol/cm ³)	$6.90 \cdot 10^{-4}$	$5.75 \cdot 10^{-4}$	$2.04 \cdot 10^{-3}$	$2.63 \cdot 10^{-4}$
ppm H ₂ O	2534	2114	7580	983
D (m ² /s)	$1.25 \pm 0.25 \times 10^{-17}$	$1.0 \pm 0.2 \times 10^{-17}$	$1.0 \pm 0.25 \times 10^{-18}$	$1.0 \pm 0.2 \times 10^{-20}$

a is the rate of retreat of the fluid/solid interface corresponding to the rate of progression of the fresh/alterned glass interface. C_1 is the hydrogen concentration measured in the centre of the pristine glass considered not to be hydrated, C_2 is the maximum hydrogen concentration measured at the fluid/solid interface, D is the apparent diffusion coefficient of hydrogen into the glass. Accuracy on water content (C_1 and C_2) is better than 20%.

alterned glass interface (Fig. 1). Analyses are performed with a constant step of 50–55 μm for potassic samples (DB, DV, RVF) and 15 μm for the sodic one (RBa). Because of the very short distance of water diffusion in the RBa sample ($\approx 60 \mu\text{m}$), only 4 analyses were carried out for this glass. All measurements were adjusted with calibration curves from [22] and obtained from the quantification of water in basaltic liquids (MORB). The detection threshold for water with these analytical conditions is better than 300 ppm. Taking into account the main sources of uncertainty [22], the estimated accuracy obtained for H₂O contents is better than $\pm 20\%$. Considering a profile performed in a same glass with the same matrix properties, the error due to the calibration is uniform for each point. Thus, the statistical error due to ion counting ($< 5\%$) has to be taken into account to validate the variations observed for a same profile. The symmetrical shape of the profile obtained for the DB sample (Figs. 2 and 3) shows the reproducibility of the analysis.

4. Results

Medieval stained-glasses submitted to damp soil for periods of time reaching up to more than one millennium

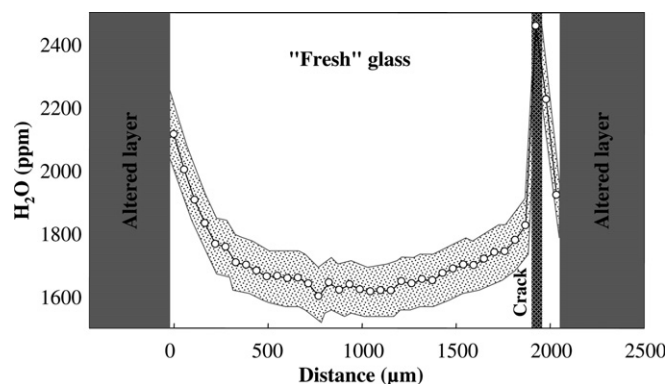


Fig. 2. H₂O concentration analysed by ion microprobe in the pristine glass from a cross-section of DB sample. Error on absolute value of water content is better than 20%. Statistical error for each point is better than 5% and is represented by the grey zone around the profile.

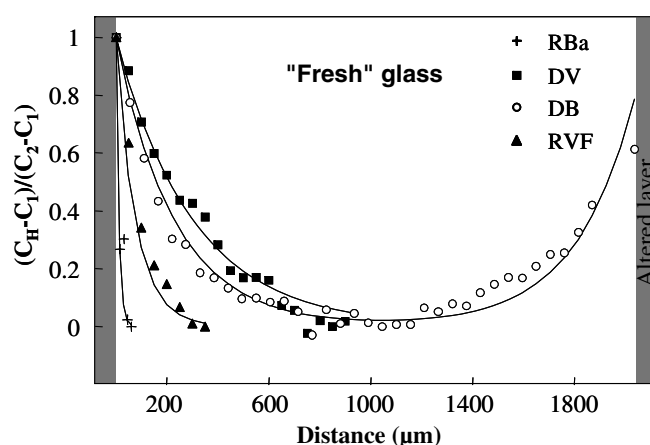


Fig. 3. Normalized hydrogen concentrations as a function of the distance in the pristine glass from the visible interface between fresh and alterned glass. The full lines correspond to the calculation from the diffusion equation (see text). The statistical error is better than 5%.

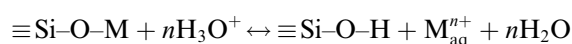
develop an alterned layer on their surface [23]. This alterned layer mainly consists of a hydrated silica gel in which some minerals such as oxides or phosphates can precipitate. Due to the hydration and very low alkali content of the alterned layer, SEM observation allows an easy distinction between this hydrated layer and the pristine glass. This simple criterion was used to define the extent of the apparent pristine glass in order to perform water analysis only into this fresh part and to avoid water contamination from the alterned layer. As an example, the diffusion profile recorded in sample DB which was analysed from one side to the other is presented in Fig. 2. The profile has a shape characteristic of a diffusion process. Maximum water concentration in the 'pristine' glass is found at the two visible interfaces between 'fresh' and alterned glass. Water content then decreases when going toward the centre of the glass showing that water comes from both sides. The four samples studied show the same behavior (Fig. 3). The water content ranges from ≈ 2100 to 7600 ppm H₂O close to the interface and from ≈ 1600 to 2800 ppm H₂O for the centre of the potassic glasses (Table 2) depending on the sample considered. The sodic, the alteration of which is less extensive, has

a water content of ≈ 1000 ppm close to the interface against ≈ 500 ppm in its centre. H_2O profiles for the four samples are plotted in Fig. 3. The water content is normalized as a function of maximum and minimum H_2O values in order to better compare the different samples and to facilitate the modelling. The normalization parameters (C_1 and C_2) are reported in Table 2. Fig. 3 shows that the depth of water penetration varies with considered samples from 50 μm for the sodic glass to 1100 μm for the less polymerized glass DB.

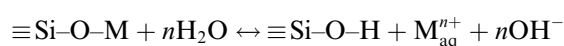
It is noteworthy that water content recorded in the fresh part of medieval stained glasses submitted to rain, i.e. submitted to a less drastic alteration, have water content of ≈ 500 ppm H_2O attributed to water trapped during glass confection [17]. Thus, lowest values are measured in the less altered glasses and could indicate that higher values could correspond not to water trapped during glass confection but to deep hydration of the glass. In other words, DB, DV and RVF samples could be fully hydrated due to weathering. For the modelling purpose we assume that minimum water content recorded in the ‘pristine’ glass corresponds to initial water before alteration. This assumption is supported by the fact that (i) real initial water content cannot be exactly known and can vary from one glass to another due to the conditions of confection, (ii) diffusion model fits better the data with this assumption rather than considering lower initial water content, (iii) RVF sample seems to show that initial water could be high since high water content in the centre of the sample was recorded (2800 ppm) despite the weaker depth of diffusion.

5. Discussion

The mechanisms responsible for hydrogen or water penetration into silicate glasses have been extensively studied (e.g. [1,9,15,24–26]). On a chemical point of view, water in contact with the free surface of the glass enters in the fresh glass by diffusion into the ionic porosity of the glass, and/or by an ionic exchange with leachable cations of the glass according to the following reaction where M is a network modifying cation (mainly alkali or alkali-earth elements):



or:



as a function of the major diffusion species, H_2O or H_3O^+ [27]. However, determination of the mechanism is difficult because of the relatively low amount of water recorded in the fresh glass (7600 ppm max) compared to alkali content (up to 18 wt% of K_2O). In this case, even if alkalis are exchanged with hydronium ions or water molecules, the amount of alkalis removed will not be sufficient to be measurable and still remain in the uncertainties of the analysis. For [9], glass alteration is due to three simultaneous reac-

tions: (1) a glass hydration due to water penetration, (2) a hydrolysis of metal–oxygen bonds and (3) an ionic exchange reaction.

Ion microprobe analyses do not allow to determine the speciation of water incorporated into the fresh glass (molecular H_2O , H_3O^+ , H^+ , silanols $Si-OH$, etc.). In this context, the stoichiometry of the reaction cannot be exactly determined, preventing to perform an exact model of water diffusion.

A simple model of water diffusion can be used to first estimate the diffusion coefficient of water into the pristine glass. Assuming that (i) hydrogen is the major species of water diffusing into the glass due to a gradient of concentration, (ii) no chemical reaction occurs between glass and water, (iii) hydration is now in a steady state, (iv) the diffusion coefficient is constant and (v) the diffusion occurs in a semi-infinite medium with a mobile boundary, the general diffusion equation is given by [1]:

$$\frac{\partial C}{\partial t} = \frac{\partial}{\partial x} \left(D \frac{\partial C}{\partial x} \right) + a \left(\frac{\partial C}{\partial x} \right)$$

with

$$a = (x' - x)/t$$

where a is the rate of retreat of the fluid/solid interface corresponding in our case to the mean rate of growth of altered layer, x' is the distance between an arbitrary plane in the altered zone and the initial fluid/solid interface (at $t = 0$), x is the distance of diffusion between this plan and the actual fluid/solid interface, C is the concentration of the diffusing species and t the time of diffusion. Solving this equation by assuming that diffusion profiles are invariant with time ($\partial C / \partial t = 0$) leads to

$$C_H = \exp \left(\frac{-ax}{D} \right)$$

where H is the diffusing hydrogen. If hydrogen exchanges with a modifying network cation M, the concentration of M as a function of the distance in the glass is given by

$$C_M = 1 - \exp \left(\frac{-ax}{D} \right)$$

assuming that H and M have the same diffusion coefficient. Because of the limited amount of water diffusing in the ‘pristine’ glass, such a profile is unfortunately not measurable. The hydrogen concentrations used in the model are normalized assuming $C_H = 1$ for $x = 0$ and $C_H = 0$ for $x \rightarrow \infty$. Table 2 contains the different parameters used to adjust the model to experimental profiles. It should be noted that water concentration C_1 corresponds to the initial water content coming from glass confection. The normalization used allows to subtract this contribution. Simulated curves compared to experimental data are plotted in Fig. 3. The diffusion coefficients were calculated by minimizing the deviation between experimental and calculated points, leading to a calculation error below 10%. Calculated apparent diffusion coefficients range from 9.9×10^{-21} to 1.2×10^{-17} m²/s, which is in agreement with

diffusion coefficients described in the case of glass alteration (e.g. [26]). The lowest value was measured for RBa sample which is the most polymerized glass.

Mean kinetics of alteration [20] as well as mineralogy of the altered layer [16] are comparable whatever the site of alteration, Rouen or Digne, showing that alteration conditions can be considered as equivalent. In this case, the only free parameter explaining the different diffusion coefficients is the chemical composition and thus the structure of glasses.

In order to assess if the diffusion coefficient of water is linked to the glass structure, apparent diffusion coefficients D_H were plotted against the bulk polymerization of the glass (NBO/T). The very good correlation, despite the low number of samples analysed, shows that the more the glass is polymerized, the less the water can penetrate into the glass structure (Fig. 4). A simple linear correlation based on our data gives an equation describing this dependence for alkali–lime–silica glasses altered in burial conditions at low temperature:

$$\text{Log}(D_H) = 4.61 \times \frac{\text{NBO}}{T} - 22.69$$

where D_H is in m^2/s . This relation gives for a total polymerization of the glass (NBO/T = 0), a diffusion coefficient D_H equal to $2.04 \times 10^{-23} \text{ m}^2/\text{s}$, independent of the nature of the glass network forming cations [28] has calculated hydration rates for obsidian. Assuming a mean temperature of 10 °C for the burial conditions, the hydration rate for obsidian, whose NBO/T is close to zero, is between 3×10^{-23} and $7 \times 10^{-23} \text{ m}^2/\text{s}$, which is in good agreement with the values deduced from medieval stained-glasses. This correlation gives direct information on the mechanisms responsible for glass hydration. Because more polymerized glasses have a more open structure due to the

adjustment of tetrahedra of network forming cations, glass hydration is not due to a simple permeation in the nanoporosity of the glass, in which case polymerized glasses should be the most hydrated ones. However depolymerization of the glass is due to the presence of ionic metal–oxygen (M–O) bonds which are more easily breakable than covalent Si–O–Si bonds, supporting the idea that glass hydration is mainly due to H–M exchange. The presence of water deep in the ‘pristine’ glass shows also that glass alteration not only begins with surface reactions. The organization of network modifying cations such as Na, K or Ca in sub-networks or regions (e.g. [29,30]) should play an important role, favouring water penetration and thus explaining the important depth of diffusion measured.

6. Conclusion

The study of buried medieval stained glasses allows the quantification of glass alteration under natural conditions over periods of time of up to one millennium. The analysis of hydrogen in pristine glass shows that (i) glass alteration is not only due to surface reactions because water penetrates into the pristine glass beyond the visible interface between pristine and altered glass, (ii) the depth of water diffusion is a function of glass polymerization : the more a glass is polymerized, the less water can enter deeply into its structure, (iii) hydrogen-cation exchange seems to be the major mechanism of glass hydration rather than a simple permeation in the nanoporosity of the glass.

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References

- [1] R.H. Doremus, *J. Non-Cryst. Solids* 19 (1975) 137.
- [2] A.F. White, H.C. Claassen, *Chem. Geol.* 28 (1980) 91.
- [3] J.O. Isard, A.R. Allnatt, P.J. Melling, *Phys. Chem. Glasses* 23 (1982) 185.
- [4] A.C. Lasaga, *J. Geophys. Res.* 89 (1984) 4009.
- [5] C. Guy, J. Schott, *Chem. Geol.* 78 (1989) 181.
- [6] M.J. Eick, P.R. Grossl, D.C. Golden, D.L. Sparks, D.W. Ming, *Geochim. Cosmochim. Acta* 60 (1996) 157.
- [7] E. Vernaz, S. Gin, C. Jégou, I. Ribet, J. Nucl. Mater. 298 (2001) 27.
- [8] T. Advocat, P. Jollivet, J.L. Crovisier, M. Del Nero, *J. Nucl. Mater.* 298 (2001) 55.
- [9] B.C. Bunker, *J. Non-Cryst. Solids* 179 (1994) 300.
- [10] J.C. Touray, *La dissolution des minéraux. Aspects cinétiques.*, 1982, 109p.
- [11] J.H. Thomassin, Ph.D Thesis, Université d'Orléans. Etude expérimentale de l'altération des verres silicatés dans l'eau douce et en milieu océanique. Apport des techniques d'analyse de surface des solides, 1984, 215p.

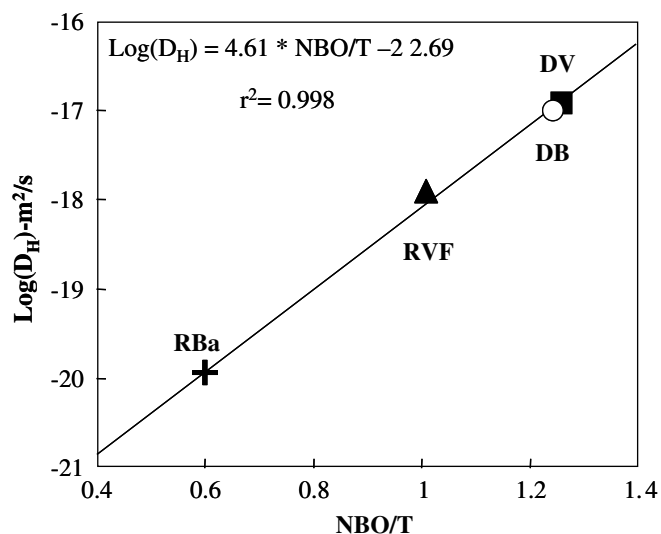


Fig. 4. Logarithm of water diffusion coefficient as a function of bulk degree of polymerization of four archaeological stained-glass samples. Error bars are smaller than symbol size.

- [12] Z. Boksay, G. Bouquet, S. Dobos, *Phys. Chem. Glasses* 8 (4) (1967) 140.
- [13] L.L. Hench, D.E. Clark, *J. Non-Cryst. Solids* 28 (1978) 83.
- [14] B.C. Bunker, T.J. Headley, S.C. Douglas, *Mat. Res. Soc. Symp. Proc.* 32 (1984) 226.
- [15] J.C. Petit, J.C. Dran, A. Paccagnella, G. Della Mea, *Earth Planet. Sci. Lett.* 93 (1989) 292.
- [16] J. Sterpenich, *Bull. Eng. Geol. Environ.* 61 (2002) 179.
- [17] J. Sterpenich, PhD Thesis, UHP Nancy 1. Altération des vitraux médiévaux. Contribution à l'étude du comportement à long terme des verres de confinement. (1998) 461p. Available from: http://tel.ccsd.cnrs.fr/documents/archives0/00/00/76/65/index_fr.html.
- [18] B.O. Mysen, in: W.S. Fyfe (Ed.), *Developments in Geochemistry*, Elsevier Science, 1988, p. 354.
- [19] M.J. Toplis, T. Schaller, *J. Non-Cryst. Solids* 224 (1998) 57.
- [20] J. Sterpenich, G. Libourel, *Chem. Geol.* 174 (2001) 181.
- [21] E. Deloule, O. Paillat, M. Pichavant, B. Scaillet, *Chem. Geol.* 125 (1995) 19.
- [22] A.V. Sobolev, M. Chaussidon, *Earth Planet. Sci. Lett.* 137 (1996) 45.
- [23] R.G. Newton, *Archeometry* 13 (1971) 1.
- [24] W. Haller, *Phys. Chem. Glasses* 4 (1963) 217.
- [25] B.M.J. Smets, T.P.A. Lommen, *Verres et réfractaires* 35 (1981) 84.
- [26] B. Grambow, R. Müller, *J. Nucl. Mater.* 298 (2001) 112.
- [27] L.R. Pederson, D.R. Baer, G.L. McVay, M.H. Engelhard, *J. Non-Cryst. Solids* 86 (1986) 369.
- [28] I. Friedman, W. Long, *Science* 191 (1976) 347.
- [29] G.N. Greaves, A. Fontaine, P. Lagarde, D. Raoux, S.J. Gurman, *Nature* 293 (1981) 611.
- [30] L. Cormier, D.R. Neuville, *Chem. Geol.* 213 (2004) 103.